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年级专业:	24 级软件工程
姓 名:	傅松煌
学 号:	2412114419
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动作识别何去何从？一个新模型与 Kinetics 数据集

作者： João Carreira, Andrew Zisserman

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摘要

当前，人类动作识别领域的最新成果主要得益于两个因素：（1）能够在大型视频数据集上进行端到端训练的深度卷积神经网络架构；（2）能够创建足够大规模视频数据集以支持这种训练方法。然而，对于第一个因素，文献中仍存在多种竞争性的网络架构设计，这些架构在性能和效率方面各有优劣；对于第二个因素，现有的视频数据集要么规模太小，要么标注质量不够精细。

本文对现有架构做了两个贡献。首先，我们仔细研究了一些因素是如何影响性能与效率之间的权衡的。这些因素包括：输入的模态（RGB、光流、RGB 差分等）、输入的视频片段时长、网络架构的选择（2D 卷积网络、3D 卷积网络、双流网络及其变体）。基于这些研究，我们得到了在测试时能够产生最高准确率同时保持合理计算开销的架构。

其次，我们引入了一个新的视频数据集——Kinetics，该数据集包含大约 30 万个视频片段，涵盖 400 种人类动作类别，每个片段由人工标注。该数据集的规模大约是现有最大视频动作识别数据集（如 UCF101 和 HMDB51）的 2 个数量级。我们在 Kinetics 上训练了当前最先进的动作识别架构，并迁移学习到 HMDB51 和 UCF101 数据集上，获得了显著的性能提升（分别提高了 10-20 个百分点）。

本文的核心贡献是提出了一个新的卷积神经网络架构——I3D（Inflated 3D ConvNet），该架构基于 2D 卷积网络 Inception v1 进行“膨胀”，将所有滤波器和池化核扩展到 3D。一个关键的发现是：通过将 2D 卷积网络在 ImageNet 上预训练的权重“膨胀”到 3D 卷积核，可以实现非常有效的迁移学习——即使对于 3D 卷积网络也是如此。因此，该模型能够在有限训练数据的情况下获得很好的性能。

我们在 Kinetics 上从头开始训练 I3D 模型，并与其他先进的动作识别架构（如 C3D、LSTM、Two-Stream CNN 等）进行了系统性的比较实验。结果表明，I3D 模型在 Kinetics 测试集上的 Top-1 准确率达到 71.1%，Top-5 准确率达到 89.3%，显著优于其他方法。更重要的是，当将在 Kinetics 上预训练的 I3D 模型迁移到 UCF101 和 HMDB51 时，准确率分别达到了 97.9% 和 80.2%，创造了当时的新纪录。

1. 引言

人类动作识别是计算机视觉领域的一个核心问题，其目标是识别视频中正在发生的人类动作类别。这一技术在视频监控、人机交互、视频内容分析、体育视频理解等领域具有广泛的应用价值。然而，动作识别是一个极具挑战性的任务，因为视频中的动作表现具有高度的多样性和复杂性——同一个人可以有不同的执行方式，不同的视角、光照条件、背景环境都会显著影响外观特征，而动作的时序变化更是增加了识别的难度。

与图像分类任务相比，视频动作识别面临额外的挑战：不仅需要分析空间外观信息，还必须理解时序动态变化。传统的动作识别方法主要依赖手工设计的特征描述符，如 HOG (Histogram of Oriented Gradients, 方向梯度直方图)、HOF (Histogram of Optical Flow, 光流直方图)、MBH (Motion Boundary Histograms, 运动边界直方图) 等。这些特征虽然在某些数据集上取得了不错的性能，但它们的设计依赖于专家经验，难以捕捉复杂的动作模式，泛化能力有限。

近年来，深度学习技术的兴起彻底改变了这一领域的研究范式。深度卷积神经网络 (Convolutional Neural Networks, CNN) 能够自动学习数据的层次化特征表示，避免了手工设计特征的局限性。然而，将 CNN 应用于视频动作识别任务并非易事。一个关键问题是如何有效地对视频数据进行建模：是将视频视为一系列独立的图像帧分别处理，还是将视频作为一个整体的 3D 数据（两个空间维度加一个时间维度）进行处理？

早期的尝试主要基于 2D CNN 架构。Karpathy 等人在 2014 年提出了使用 2D CNN 直接处理视频帧的方法，他们探索了多种融合策略（早期融合、晚期融合、慢融合），试图将时序信息整合到网络中。然而，这种方法的性能并不理想，主要原因是 2D 卷积只能在空间维度上提取特征，无法直接建模时序动态。

Simonyan 和 Zisserman 在 2014 年提出的双流卷积网络 (Two-Stream Convolutional Networks) 是一个重要的突破。该方法包含两个独立的网络分支：空间流 (Spatial Stream) 使用 2D CNN 处理单帧 RGB 图像，捕获外观信息；时间流 (Temporal Stream) 使用 2D CNN 处理密集光流图像，捕获运动信息。两个网络的预测结果通过后期融合得到最终分类。这一方法在 UCF101 和 HMDB51 数据集上取得了显著的性能提升，证明了显式建模运动信息的重要性。

然而，双流网络的一个主要缺点是需要预先计算光流，这是一项计算密集型的预处理步骤。为了解决这个问题，Tran 等人在 2015 年提出了 C3D (3D Convolutional Neural Networks) 模型，使用 3D 卷积核直接在视频数据上进行操作，同时学习时空特征。3D 卷积的核是 4 维的（宽度、高度、时间、通道数），可以在时空域上同时进行卷积操作。C3D 的优势在于无需预先计算光流，可以直接端到端地训练。然而，由于 3D 卷积的参数数量远大于 2D 卷积，C3D 面临着过拟合的风险，尤其是在训练数据有限的情况下。

在本文中，我们提出了一个新的 3D 卷积网络架构——I3D (Inflated 3D ConvNet)，旨在解决上述方法的局限性。I3D 的核心思想是：利用在大型图像数据集（如 ImageNet）上预训练的 2D 卷积网络的权重，通过“膨胀”操作将其转换为 3D 卷积权重。具体来说，我们将 2D 卷积核沿着时间维度重复复制，形成一个 3D 卷积核。例如，一个 $N \times N$ 的 2D 卷积核被膨胀为一个 $N \times N \times N$ 的 3D 卷积核。这种方法的优势在于：（1）可以利用 ImageNet 预训练模型的强大特征学习能力；（2）膨胀后的 3D 卷积核具有合理的时间感受野初始化；（3）通过微调，网络可以快速适应视频数据。

为了训练和评估 I3D 模型，我们还引入了一个新的大规模视频动作识别数据集——Kinetics。现有的视频动作识别数据集（如 UCF101 包含 13,320 个视频，HMDB51 包含 6,766 个视频）规模相对较小，限制了深度神经网络的性能。Kinetics 数据集包含大约 300,000 个视频片段，涵盖 400 种动作类别，每个视频片段约 10 秒，由人工标注。该数据集的规模是现有数据集的约 20 倍，为训练深度 3D 卷积网络提供了充足的数据支持。

我们的实验表明，在 Kinetics 数据集上从头开始训练的 I3D 模型，性能显著优于其他先进的动作识别架构。更重要的是，当将在 Kinetics 上预训练的 I3D 模型迁移到其他较小的数据集（如 UCF101 和 HMDB51）时，获得了显著的性能提升，创造了当时的新纪录。这一结果证明了大规模数据集和有效迁移学习策略对于视频动作识别任务的重要性。

本文的结构安排如下：第 2 节详细介绍了 I3D 模型的架构设计；第 3 节描述了 Kinetics 数据集的构建过程；第 4 节介绍了实验设置和结果分析；第 5 节总结了本文的主要贡献，并讨论了未来的研究方向。

2. I3D 模型架构

I3D 模型基于 Inception v1 架构进行膨胀。Inception v1 是一种经典的 2D 卷积网络架构，其核心思想是使用不同尺度的卷积核并行处理输入，然后将结果拼接起来。我们将 Inception v1 中的所有 2D 卷积核和 2D 池化核膨胀为 3D 卷积核和 3D 池化核。

具体来说，对于一个 $N \times N$ 的 2D 卷积核，我们将其膨胀为一个 $N \times N \times N$ 的 3D 卷积核，即将原始权重沿着时间维度复制 N 次。这种膨胀策略确保了膨胀后的 3D 卷积核在时间维度上具有合理的行为：网络最初对待时间维度与空间维度是等同的，但通过后续的训练微调，网络可以学习到时空特征的最佳表示方式。

在输入处理方面，I3D 模型接受可变长度的视频片段作为输入。为了捕获长期时序信息，我们通常输入包含 64 帧的连续视频片段。网络的第一层是一个时空卷积层，使用 $7 \times 7 \times 7$ 的卷积核，步长为 $2 \times 2 \times 2$ 。随后是多个 Inception 模块，每个模块包含并行的不同尺度的 3D 卷积核。最后是全局平均池化层和全连接分类层。

需要注意的是，我们在膨胀过程中对所有卷积核使用相同的膨胀比例（沿时间维度复制 N 次）。这种设计保持了网络的简洁性，实验表明这种简单的膨胀策略在实践中效果良好。当然，也可以考虑使用不同的膨胀比例（例如，在空间维度上使用 $N \times N$ ，在时间维度上使用 M ），但这会增加超参数的选择难度，我们将其留给未来的工作。

3. Kinetics 数据集

Kinetics 是一个大规模的视频动作识别数据集，旨在为深度学习模型的训练提供充足的数据支持。数据集的构建过程如下：

首先，我们定义了 400 种人类动作类别，涵盖了人与物体交互（如演奏乐器）、人体动作（如握手）、人与人交互（如拥抱）等多种类型。每个类别包含 400 到 1,000 个视频片段。

视频片段来源于 YouTube，每个片段约 10 秒，由人工标注。为了确保标注质量，我们采用了多轮标注和质量检查机制。每个视频片段由至少两名标注员进行标注，只有当标注结果一致时才接受该标注。

Kinetics 数据集被划分为训练集、验证集和测试集，比例约为 70%、10%、20%。测试集的标注不对外公开，研究者需要提交预测结果到在线评测服务器进行评估。

Kinetics 数据集的规模大约是现有最大视频数据集（UCF101）的 20 倍。这使得我们能够训练更深的 3D 卷积网络，而无需担心严重的过拟合问题。此外，Kinetics 的类别更加多样化，包含了许多之前数据集中未出现的动作类型，为研究更通用的动作识别模型提供了机会。

4. 实验与结果

我们在 Kinetics、UCF101 和 HMDB51 三个数据集上进行了实验，比较了 I3D 模型与其他先进方法的性能。

在 Kinetics 数据集上，I3D 模型（RGB+光流双流融合）的 Top-1 准确率达到 71.1%，Top-5 准确率达到 89.3%，显著优于其他方法。C3D 的 Top-1 准确率为 61.1%，Two-Stream CNN 为 62.0%，LSTM 为 63.3%。

为了验证迁移学习的效果，我们将在 Kinetics 上预训练的 I3D 模型迁移到 UCF101 和 HMDB51 数据集上。结果表明，迁移学习显著提升了性能：在 UCF101 上，I3D 的准确率达到 97.9%，相比从头训练的 Two-Stream CNN（88.0%）提升了近 10 个百分点；在 HMDB51 上，I3D 的准确率达到 80.2%，相比从头训练的方法提升了约 20 个百分点。

这些结果充分证明了：（1）3D 卷积网络在视频动作识别中的有效性；（2）大规模数据集（Kinetics）对于训练深度视频模型的重要性；（3）从图像到视频的迁移学习策略的有效性。

5. 结论与展望

本文提出了 I3D 模型，一种新的 3D 卷积网络架构用于视频动作识别。I3D 的核心创新在于将 2D 卷积网络在 ImageNet 上预训练的权重“膨胀”到 3D 卷积核，实现了高效的迁移学习。同时，我们引入了 Kinetics 数据集，一个大规模的包含 400 种动作类别的视频数据集。

实验结果表明，I3D 模型在 Kinetics 上取得了最先进的性能，并且通过迁移学习，在其他数据集上也创造了新的记录。这些成果推动了视频动作识别领域的研究进展。

未来的研究方向包括：（1）探索更高效的 3D 卷积架构，减少计算开销；（2）研究无监督或半监督学习方法，减少对标注数据的依赖；（3）将动作识别技术应用于更广泛的实际场景，如视频监控、人机交互、体育分析等。我们相信，随着深度学习技术的不断发展和更大规模数据集的出现，视频动作识别技术将在更多领域发挥重要作用。

Abstract

The paucity of videos in current action classification datasets (UCF-101 and HMDB-51) has made it difficult to identify good video architectures, as most methods obtain similar performance on existing small-scale benchmarks. This paper re-evaluates state-of-the-art architectures for video action classification on two new large-scale video datasets: Kinetics-400 (400 classes and 400 clips per class), and Kinetics-600 (600 classes). Two architectures are considered: (i) inflation: extending 2D ConvNets to 3D by repeating 2D filters (and optionally pooling operators) along the time dimension, and (ii) two-stream networks. We show that an inflated 3D ConvNet (I3D) based on an Inception v1 architecture is the best performer on the Kinetics datasets. Furthermore, I3D models pre-trained on Kinetics can be fine-tuned on UCF-101 and HMDB-51 and achieve state-of-the-art results on these benchmarks: 97.9% on UCF-101 and 80.2% on HMDB-51. We also show that I3D features can be used as a generic video representation for other video analysis tasks, such as action detection and video captioning.

1. Introduction

Human action recognition is a core problem in computer vision with applications in video surveillance, human-computer interaction, video content analysis, and sports video understanding. Despite significant progress in recent years, action recognition remains a challenging task due to the high variability in how actions are performed, different viewpoints, lighting conditions, and the complex temporal dynamics involved.

Compared to image classification, action recognition has additional challenges: not only must spatial appearance information be analyzed, but temporal dynamics must also be understood. Traditional action recognition methods relied on hand-crafted feature descriptors such as HOG (Histogram of Oriented Gradients), HOF (Histogram of Optical Flow), and MBH (Motion Boundary Histograms). While these features achieved reasonable performance on some datasets, their design relies on expert experience, making it difficult to capture complex action patterns and limiting their generalization

capabilities.

The rise of deep learning has fundamentally changed the research paradigm in this field. Deep Convolutional Neural Networks (CNNs) can automatically learn hierarchical feature representations from data, avoiding the limitations of hand-crafted features. However, applying CNNs to video action recognition is not straightforward. A key question is how to effectively model video data: should videos be treated as a series of independent image frames, or as a unified 3D volume (two spatial dimensions plus one temporal dimension)?

Early attempts primarily used 2D CNN architectures. In 2014, Karpathy et al. proposed using 2D CNNs to directly process video frames, exploring multiple fusion strategies (early fusion, late fusion, slow fusion) to incorporate temporal information. However, this approach did not perform well, mainly because 2D convolutions can only extract features in spatial dimensions and cannot directly model temporal dynamics.

Simonyan and Zisserman's Two-Stream Convolutional Networks, proposed in 2014, marked an important breakthrough. This method consists of two independent network branches: the Spatial Stream uses a 2D CNN to process single RGB frames, capturing appearance information; the Temporal Stream uses a 2D CNN to process dense optical flow images, capturing motion information. The predictions from both networks are combined through late fusion to produce the final classification. This method achieved significant performance improvements on UCF-101 and HMDB-51 datasets, demonstrating the importance of explicitly modeling motion information.

However, a major drawback of two-stream networks is the need to pre-compute optical flow, which is a computationally intensive preprocessing step. To address this issue, Tran et al. proposed the C3D (3D Convolutional Neural Networks) model in 2015, which uses 3D convolutional kernels to operate directly on video data, learning spatiotemporal features simultaneously. 3D convolution kernels are 4-dimensional (width, height, time, channels), allowing convolution operations across both spatial and temporal domains. The advantage of C3D is that it does not require pre-computed optical flow and can be trained end-to-end. However, due to the significantly larger number of parameters in 3D convolutions compared to 2D convolutions, C3D faces the risk of overfitting, especially

when training data is limited.

In this paper, we propose a new 3D convolutional network architecture called I3D (Inflated 3D ConvNet) to address the limitations of existing methods. The core idea of I3D is to leverage weights from 2D convolutional networks pre-trained on large image datasets (such as ImageNet) and "inflate" them into 3D convolutional weights. Specifically, we replicate 2D convolutional kernels along the time dimension to form 3D convolutional kernels. For example, an $N \times N$ 2D convolutional kernel is inflated into an $N \times N \times N$ 3D convolutional kernel. The advantages of this approach are: (1) it leverages the powerful feature learning capabilities of ImageNet pre-trained models; (2) the inflated 3D convolutional kernels have reasonable temporal receptive field initialization; (3) through fine-tuning, the network can quickly adapt to video data.

To train and evaluate the I3D model, we also introduce a new large-scale video action recognition dataset called Kinetics. Existing video action recognition datasets (such as UCF-101 with 13,320 videos and HMDB-51 with 6,766 videos) are relatively small in scale, limiting the performance of deep neural networks. The Kinetics dataset contains approximately 300,000 video clips covering 400 action categories, with each video clip lasting about 10 seconds and annotated by humans. This dataset is about 20 times larger than existing datasets, providing sufficient data support for training deep 3D convolutional networks.

Our experiments show that I3D models trained from scratch on the Kinetics dataset significantly outperform other advanced action recognition architectures. More importantly, when transferring I3D models pre-trained on Kinetics to other smaller datasets (such as UCF-101 and HMDB-51), we achieve significant performance improvements, setting new records at the time. These results demonstrate the importance of large-scale datasets and effective transfer learning strategies for video action recognition tasks.

The remainder of this paper is organized as follows: Section 2 describes the I3D model architecture in detail; Section 3 describes the construction of the Kinetics dataset; Section 4 presents experimental setup and results analysis; Section 5 summarizes the main contributions of this paper and discusses future research directions.

2. The I3D Architecture

The I3D model is based on inflating the Inception v1 architecture. Inception v1 is a classic 2D convolutional network architecture whose core idea is to process inputs using convolutional kernels of different scales in parallel, then concatenate the results. We inflate all 2D convolutional kernels and 2D pooling kernels in Inception v1 into 3D convolutional kernels and 3D pooling kernels.

Specifically, for an $N \times N$ 2D convolutional kernel, we inflate it into an $N \times N \times N$ 3D convolutional kernel by replicating the original weights N times along the time dimension. This inflation strategy ensures that the inflated 3D convolutional kernels have reasonable behavior in the temporal dimension: the network initially treats the temporal dimension equally with spatial dimensions, but through subsequent fine-tuning, the network can learn the optimal representation of spatiotemporal features.

For input processing, the I3D model accepts video clips of variable length as input. To capture long-term temporal information, we typically input continuous video clips containing 64 frames. The first layer of the network is a spatiotemporal convolutional layer using $7 \times 7 \times 7$ kernels with a stride of $2 \times 2 \times 2$. This is followed by multiple Inception modules, each containing parallel 3D convolutional kernels of different scales. Finally, there is a global average pooling layer and a fully connected classification layer.

It is worth noting that we use the same inflation ratio for all convolutional kernels during inflation (replicating N times along the time dimension). This design maintains the simplicity of the network, and experiments show that this simple inflation strategy works well in practice. Of course, different inflation ratios could be considered (for example, using $N \times N$ in spatial dimensions and M in temporal dimension), but this would increase the difficulty of hyperparameter selection, which we leave for future work.

3. The Kinetics Dataset

Kinetics is a large-scale video action recognition dataset designed to provide sufficient data support for training deep learning models. The dataset construction process is as follows:

First, we defined 400 human action categories, covering various types including human-object interaction (such as playing musical instruments), human body movements (such as shaking hands), and human-human interaction (such as hugging). Each category contains between 400 and 1,000 video clips.

Video clips are sourced from YouTube, with each clip lasting approximately 10 seconds and annotated by humans. To ensure annotation quality, we employed multi-round annotation and quality checking mechanisms. Each video clip is labeled by at least two annotators, and the annotation is only accepted when the labeling results agree.

The Kinetics dataset is divided into training, validation, and test sets with approximate ratios of 70%, 10%, and 20%. The annotations for the test set are not publicly released; researchers need to submit their prediction results to an online evaluation server for assessment.

The Kinetics dataset is approximately 20 times larger than the largest existing video dataset (UCF-101). This allows us to train deeper 3D convolutional networks without worrying about severe overfitting. Furthermore, Kinetics has more diverse categories, including many action types not present in previous datasets, providing opportunities for studying more general action recognition models.

4. Experiments and Results

We conducted experiments on three datasets: Kinetics, UCF-101, and HMDB-51, comparing the performance of the I3D model with other state-of-the-art methods.

On the Kinetics dataset, the I3D model (RGB+optical flow two-stream fusion) achieved a Top-1 accuracy of 71.1% and Top-5 accuracy of 89.3%, significantly outperforming other methods. C3D achieved 61.1% Top-1 accuracy, Two-Stream CNN 62.0%, and LSTM 63.3%.

To verify the effectiveness of transfer learning, we transferred I3D models pre-trained on Kinetics to UCF-101 and HMDB-51 datasets. The results show that transfer learning significantly improved performance: on UCF-101, I3D achieved 97.9% accuracy, nearly 10 percentage points higher than Two-Stream CNN trained from scratch (88.0%); on HMDB-51, I3D achieved 80.2% accuracy, about 20 percentage points higher than

methods trained from scratch.

These results fully demonstrate: (1) the effectiveness of 3D convolutional networks in video action recognition; (2) the importance of large-scale datasets (Kinetics) for training deep video models; (3) the effectiveness of transfer learning strategies from images to video.

5. Conclusion and Future Work

This paper proposes the I3D model, a new 3D convolutional network architecture for video action recognition. The core innovation of I3D is inflating weights from 2D convolutional networks pre-trained on ImageNet into 3D convolutional kernels, enabling efficient transfer learning. We also introduce the Kinetics dataset, a large-scale video dataset containing 400 action categories.

Experimental results show that the I3D model achieves state-of-the-art performance on Kinetics, and through transfer learning, also sets new records on other datasets. These advances push forward the research progress in video action recognition.

Future research directions include: (1) exploring more efficient 3D convolutional architectures to reduce computational overhead; (2) studying unsupervised or semi-supervised learning methods to reduce dependence on labeled data; (3) applying action recognition technology to broader practical scenarios such as video surveillance, human-computer interaction, and sports analysis. We believe that as deep learning technology continues to evolve and larger-scale datasets become available, video action recognition technology will play increasingly important roles in more fields.

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